

10. FORMATION OF THE UPPER INTESTINE

DURATION : 25:32

STUDY SHEET

COURSE SUMMARY

This video deeply explores the development of the upper intestine, highlighting the complex movements of the endoderm and vascular interactions. You will learn how these embryological processes result in the formation of the pharyngeal arches and the organisation of facial structures, and how this knowledge can be applied in a therapeutic context. The concepts of embryonic flexion, cerebralisation, and the impact of osteopathy on the dynamics of the mandibular arches and dental structures are discussed, thus offering practical tools to restore harmony in the face of internal architecture.

FORMATION OF THE UPPER INTESTINE

The formation of the upper intestine is a complex process involving the movement of the **endoderm**. This movement manifests both in its entirety and **transversely**, transforming a sac into a tube. The **amniotic cavity** plays a key role in this movement, completely enveloping the embryo and integrating the digestive tract from the outside inwards, primarily guided by the primitive **aortic tether**.

Focusing on the **vascular** plane, we observe that the primitive endocardial tubes approach each other simultaneously with the closure of the neural tube. This meeting creates a space in the lumen of the **primitive vitelline endoderm**, forming what is known as the **pharyngeal gut**. This process occurs on the midline, while laterally, the **cellulose** appears.

The differential growth of **epithelial tissue** relative to **connective tissue** allows the two aortas to come together on the central axis, thus defining the longitudinal axis and creating the pharyngeal gut. The upper intestine extends from the mouth to the mid-part, involving the formation of the face.

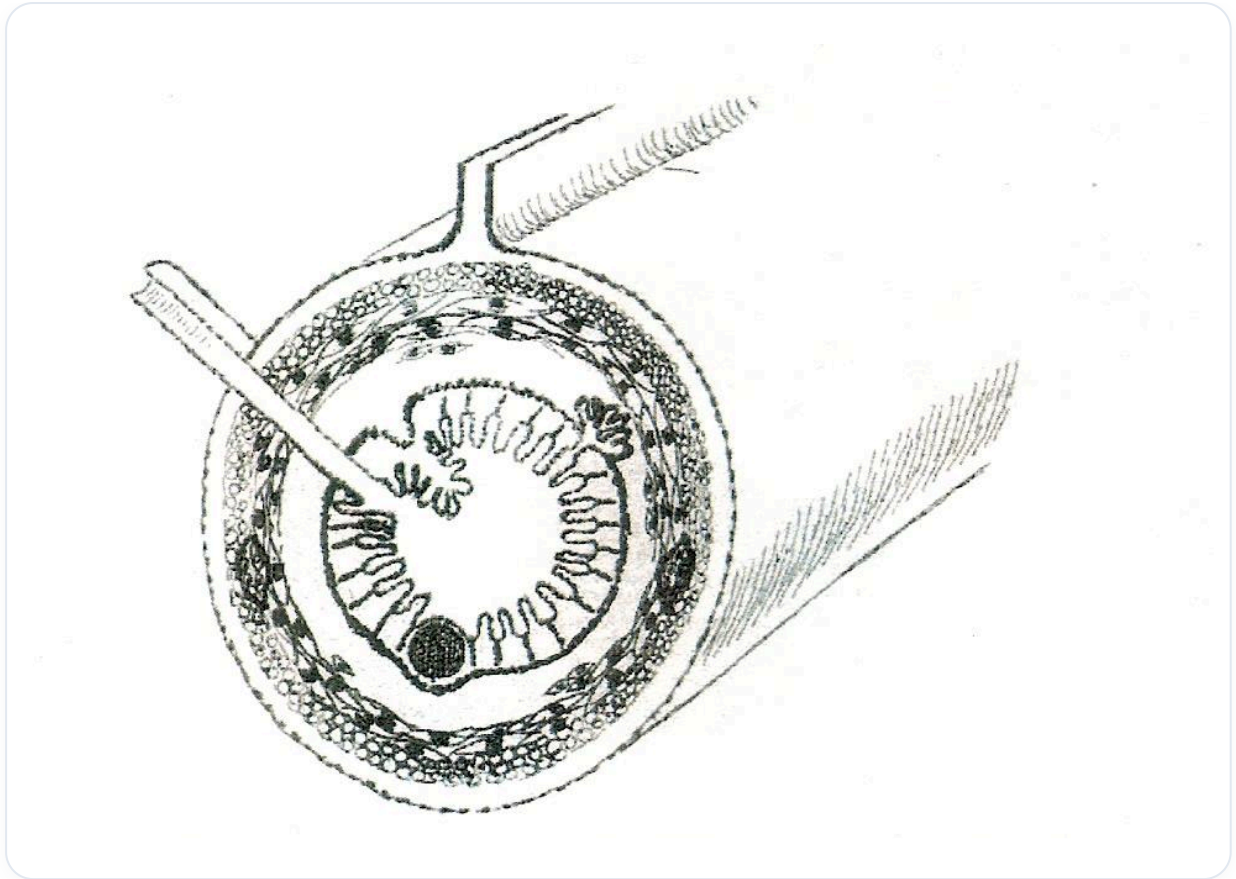


FIG. — Intestine cross-section

At approximately fifteen days, the end of the second week, the embryo can be divided into three parts: the **upper third**, the **middle third**, and the **lower third**. The head of the future embryo is disproportionate to a small trunk, with the cephalic part already identifiable. Half of the embryo is located on the base of the skull, in relation to the **notochordal field** relative to the primitive streak.

The flexion of the embryo is influenced by the vascular system. A simple diagram shows that a tube bends, with one area densifying and another stretching, thus forming **plicatures**. Internally, the tissue is **endodermal**, while externally it is **ectodermal**, with **mesodermal** tissue in between. The branchial arches, organised in space relative to the embryo, contain specific information.

Each branchial arch is oriented towards the base of the skull. The mandibular arches, for example, converge towards this base. Treating a face or a skull involves restoring this primitive function, by re-establishing an **energetic field** that organises towards this space.

Four major functional branchial arches can be identified: the **mandibular arch**, the **hyoid arch**, the **superior laryngeal arch**, and the **inferior laryngeal arch**. Each arch, whether ectodermal, mesodermal, or endodermal, contains specific information. The mandibular arch, for example, is linked to the mandibular cartilage and structures of the **inner ear**. Therefore, destabilisation of the jaw can affect the internal organisation of the ear.

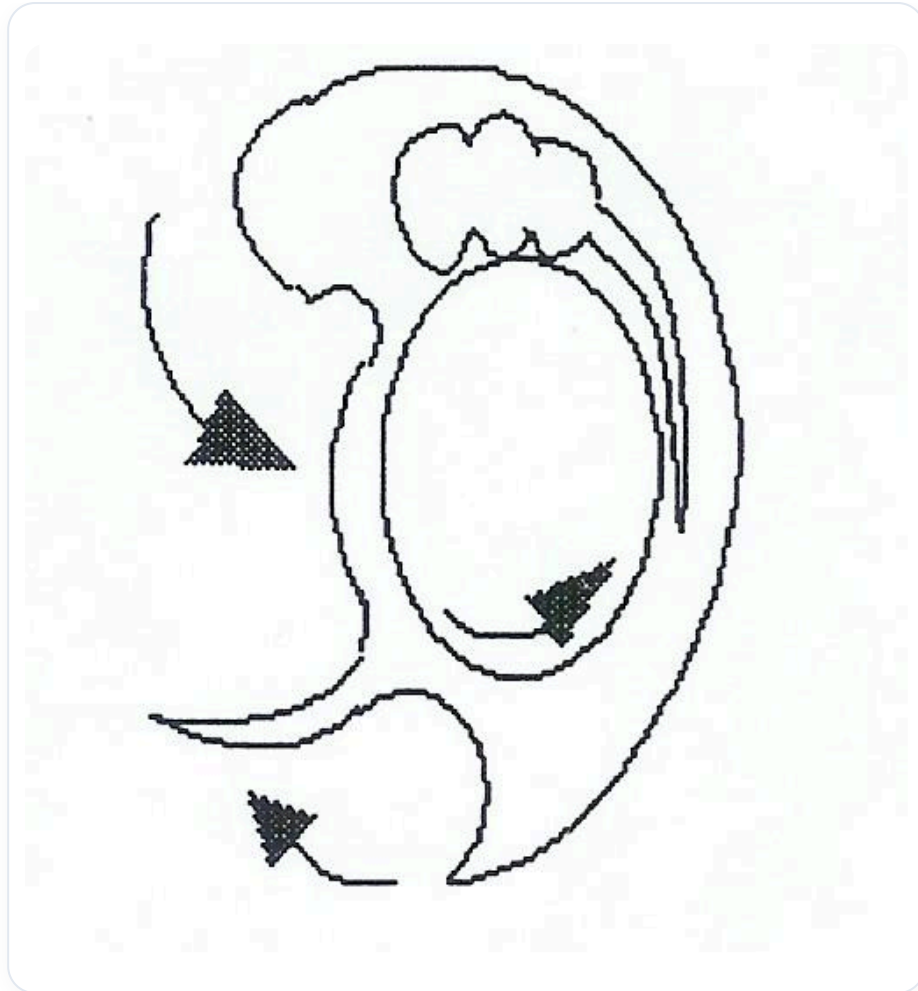


FIG. — Embryonic inner ear

The process of embryonic flexion and elongation is also linked to **cerebralisation** and the formation of the **prosencephalic system**. This development occurs in a **biodynamic** process, where the embryo begins to fold and open. The heart develops between the brain, the neural tube, and the liver, thus creating the face.

The rupture of the **buccopharyngeal membrane** marks the beginning of the formation of the mouth, which develops horizontally due to compression between the heart and the brain. This compression integrates the mesodermal tissue, containing vascular information. The appearance of the **nasal pit** and the oral cavity is also linked to this dynamic.

Brain development is related to the process of **telencephalisation**, which involves an ascent of the brain and a visceral descent. The face, initially, is a field of compression between the heart and the brain, and through the growth of the embryo, it opens.

The development of the **pterygoids** and the **palate** is influenced by brain traction and visceral descent. This synchronised movement between the digestive system and the heart, as well as cerebral ascent, contributes to the formation of the ogival palate.

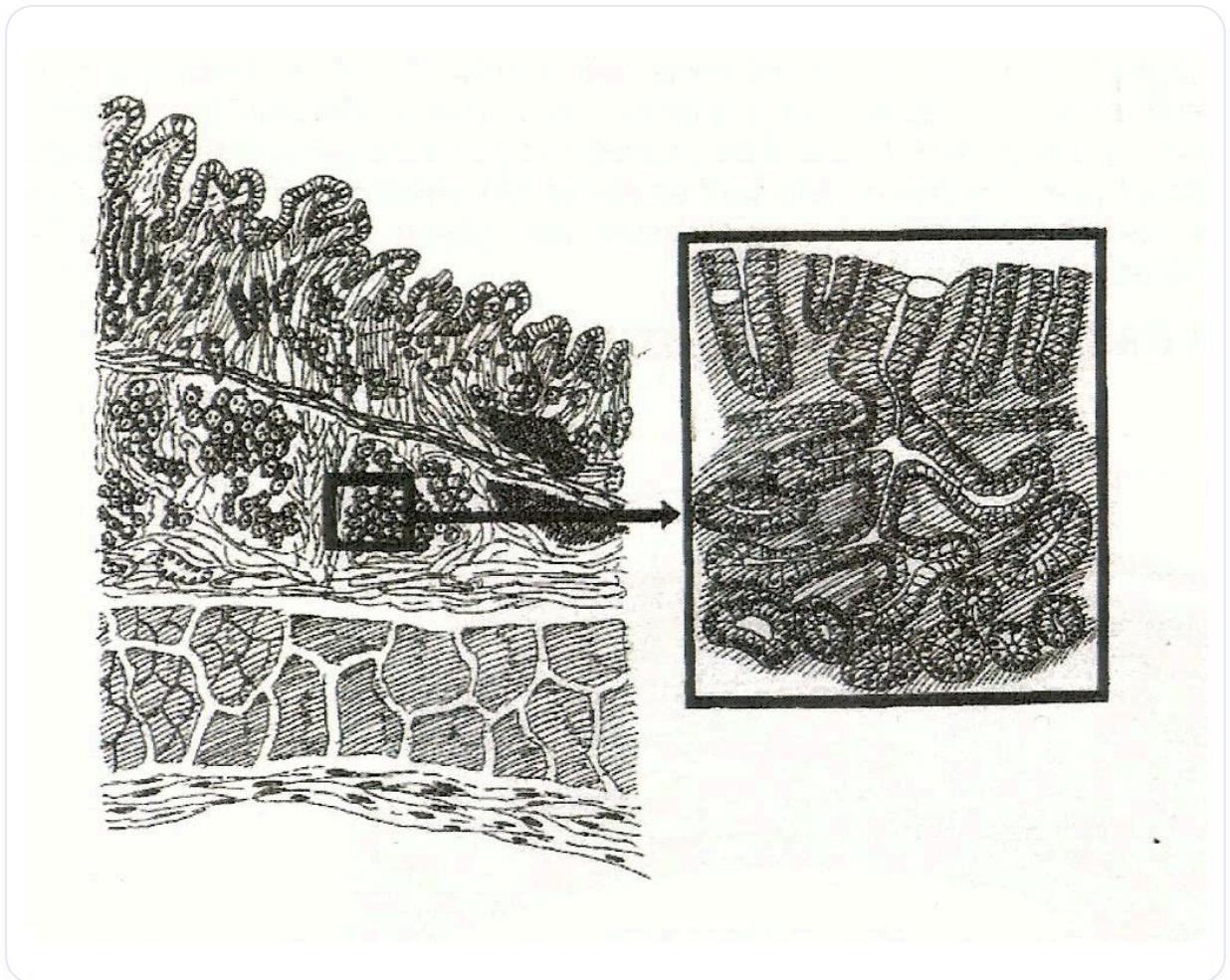


FIG. — Intestinal villi tissue

Occlusal forces, pressures from the lingual mass, and piezoelectric effects also influence the formation of teeth and facial structures. Each tooth receives specific information, and treating the face or teeth requires rebalancing these structures with their embryonic origin.

In conclusion, the formation of the upper intestine and facial structures is a complex process, involving dynamic interactions between different embryonic tissues and their synchronised development.

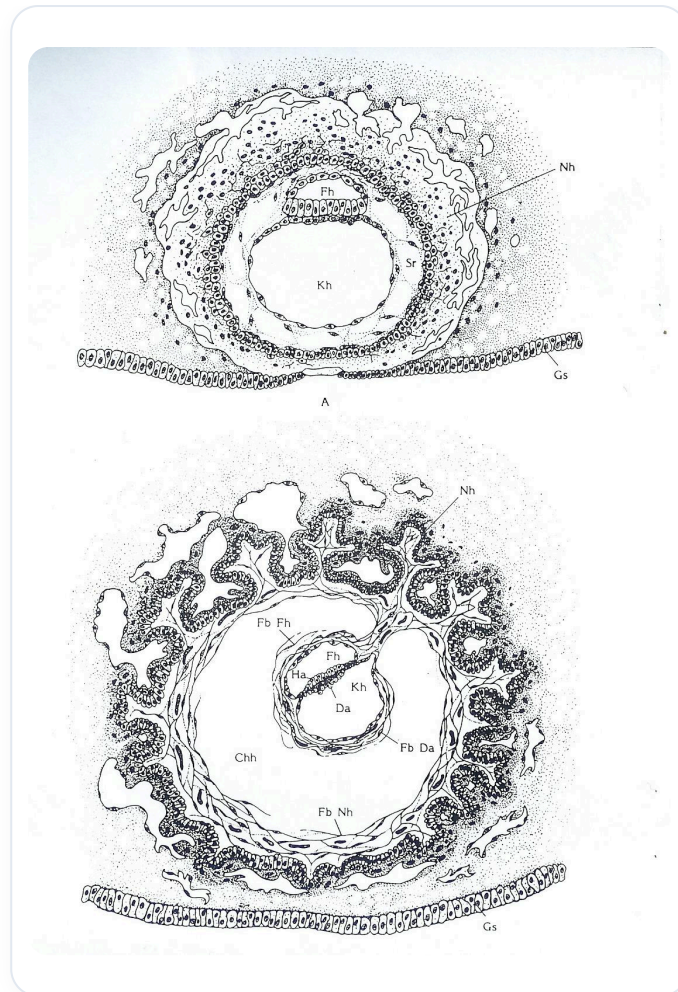


FIG. — *Bird embryonic development*

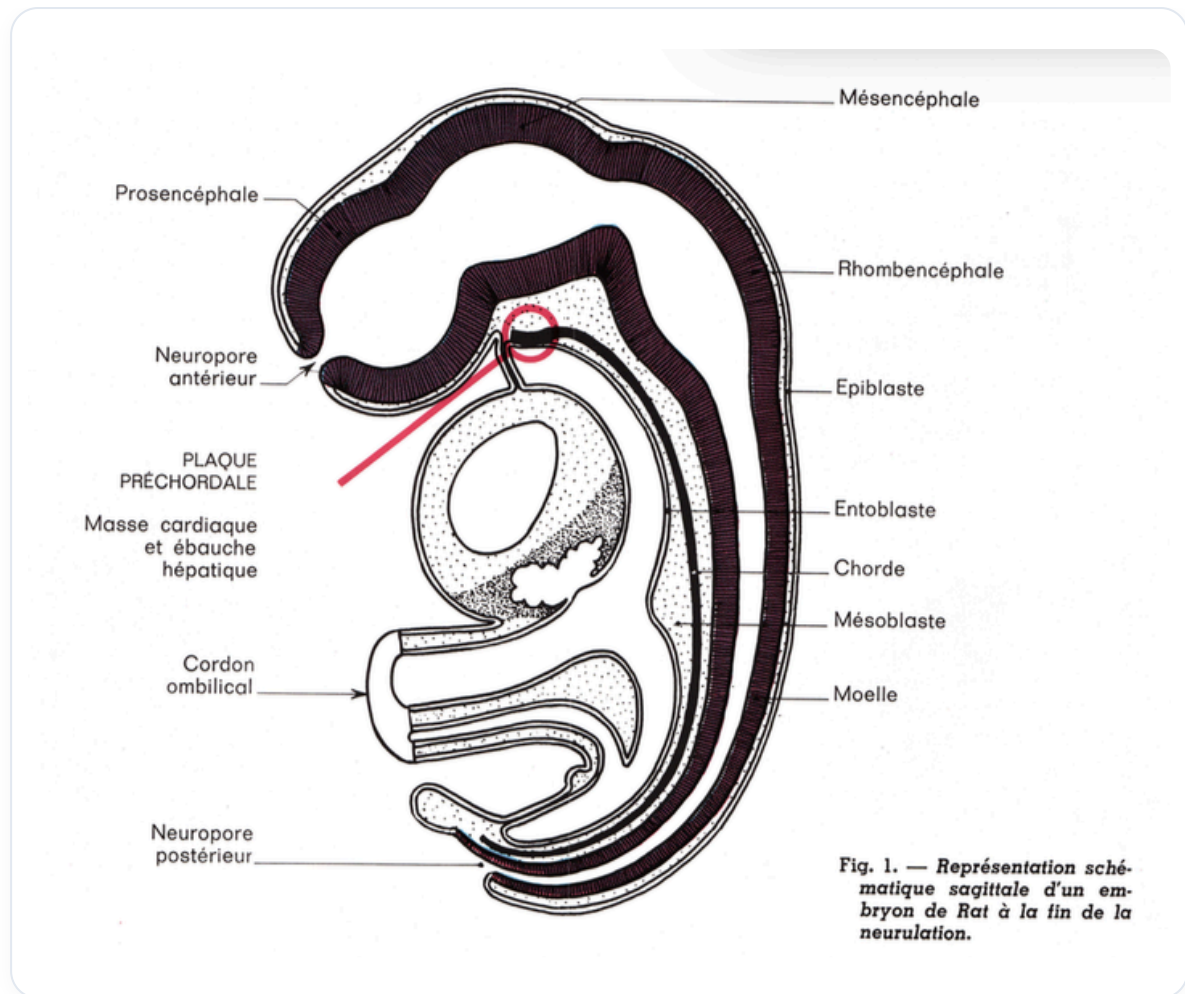


FIG. — Rat embryo neurulation

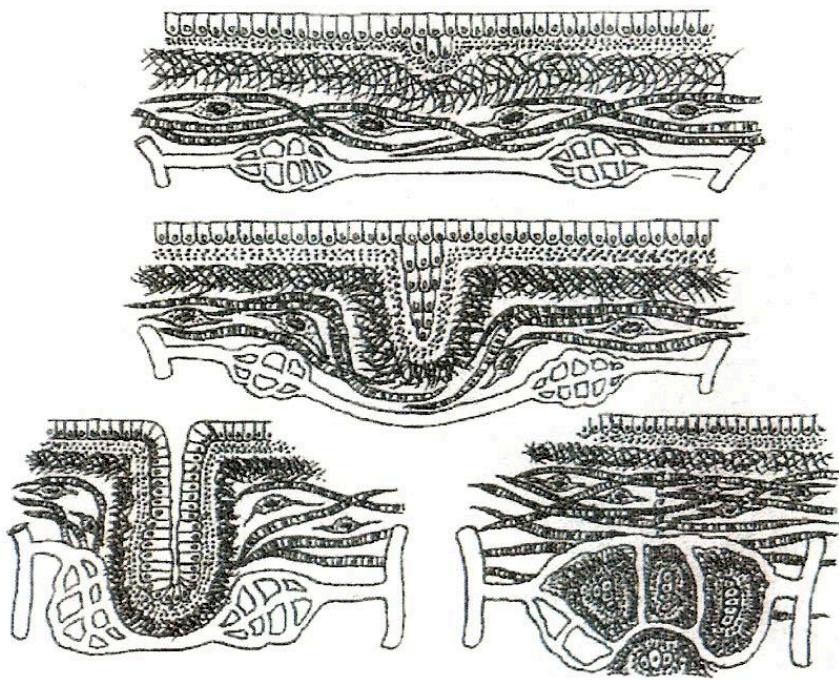


FIG. — Intestinal villification